

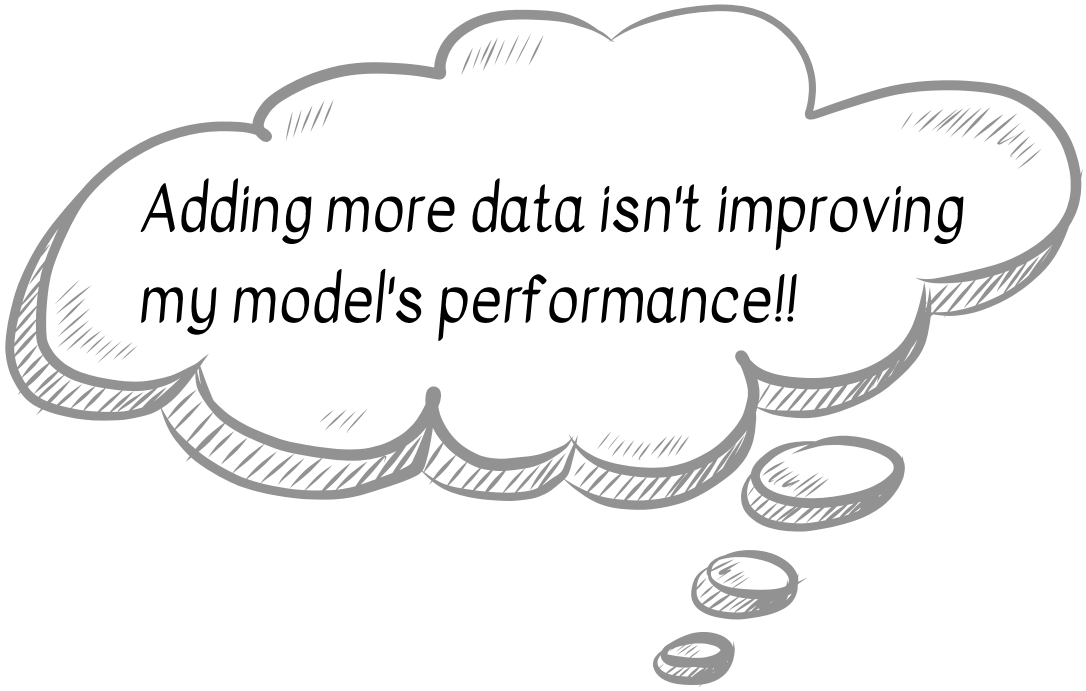
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A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .

DOES MORE DATA (EXPERIENCE) LEAD TO BETTER PERFORMANCE?





*Adding more data isn't improving
my model's performance!!*

Bias— Bias refers to the error from simplifying a real-world problem with an overly simple model. High bias can cause the model to overlook important relationships between input features and output predictions.

With high bias, adding more data might not help as the model is too basic to grasp data complexities.

Variance— Reflects the model's responsiveness to small fluctuations or noise in training data. A model with high variance excels on its training data by accounting for random noise, but falters on new, unseen data (overfitting).

With high variance, adding more data can often boost performance since it helps "average out" or reduce the noise captured by the model.



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